



Linux OS – Lab 7

Task 1: Managing Processes

1. Explain the various process states in Linux.
2. Check disk usage of specific directories using **du** and **df**.
3. Run a **sleep** command in the **foreground**.
4. Stop the running foreground process then send it to the **background**.
5. Run five **sleep** commands in the **background**.
6. List all the running jobs along with their **PID**.
7. Bring the last background process to the foreground then **terminate** it.
8. Kill specific background processes using their **PID** with **SIGKILL** and **SIGTERM** signals.
9. Find the **PID** of **vim** process and terminate it.

Task 2: Process Monitoring

1. List all the running processes in a long format using **ps** command.
2. Run the **top** utility and view active processes in real-time.
3. Limit the number of processes to be **5**.
4. Sort processes by memory usage.
5. Change the refresh interval to 0.1 seconds.
6. Save this configuration for reuse when **top** is restarted.
7. Kill a process from within **top**.
8. Change the priority of a process using renice by pressing '**r**' within **top**.
9. Run another **sleep** process but with a nice value of -5.
10. Renice the last running **sleep** process to have a nice value of 10.
11. Kill all the running background processes.